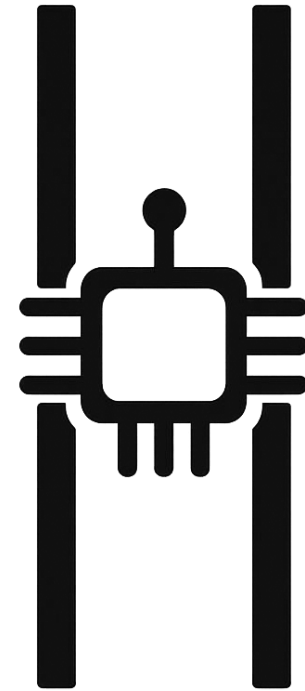

VISIONLANE

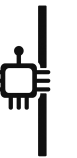
Full AI-Traffic detection/prediction



1. Problem

- Traffic rises (2% every year)
- Live feed
- Unreliable predictions (on feeling)

INCONSISTENCY



2. Solution

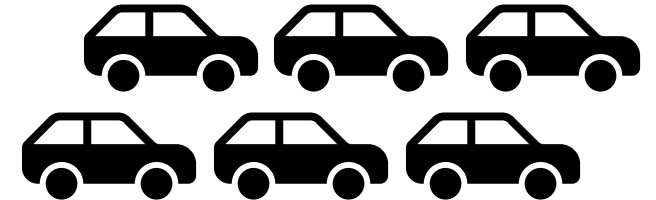
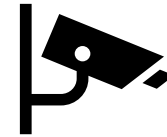
- Real time data (learns from data)
- Rush hours & night traffic
- Prediction
- "How wil the traffic be when I leave?"



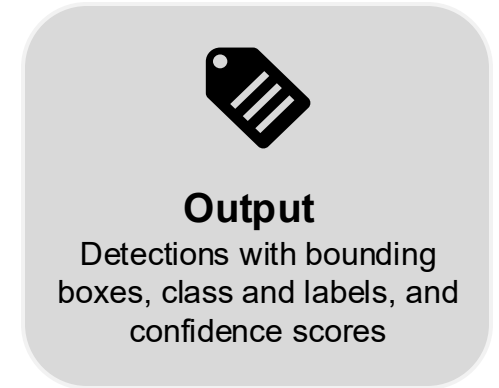
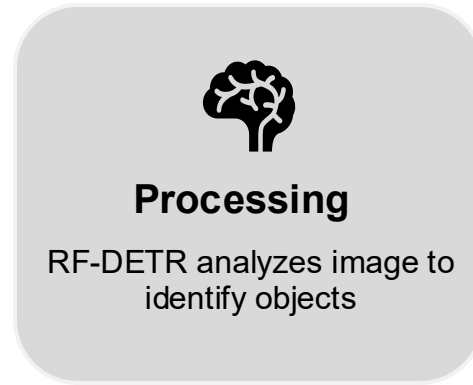
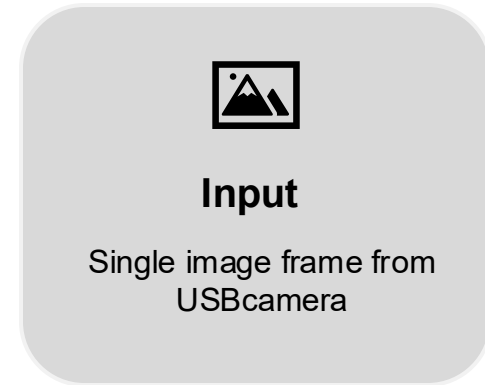
3. Approach

Data collection

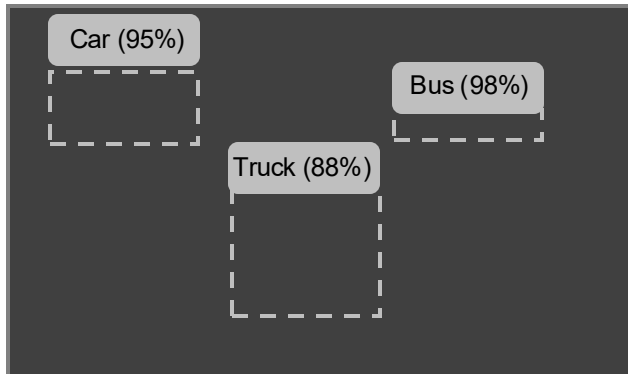
- Collect traffic information  Embedded linux board
- RF DETR model (RT, transformer-based object detection model)
- COCO dataset
- Sends data to database in the Virtual machine



3.1 Rock5



Detection Output



Vehicle Counting

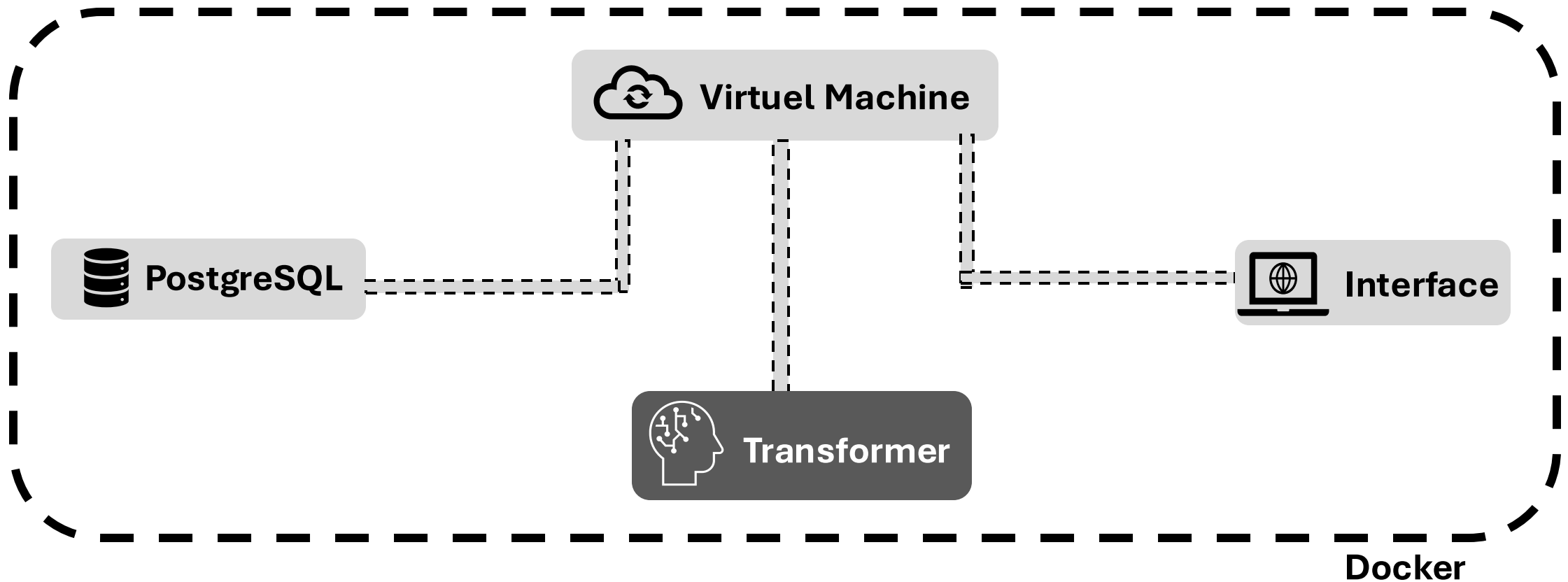


JSON Output

```
{ "timestamp": "2025-12-12T10:30:00Z",  
  "total_vehicles": 8,  
  "breakdown": {  
    "car": 5,  
    "truck": 2,  
    "bus": 1,  
    "motorcycle": 0,  
    "bicycle": 0  
  },  
  "camera_id": "camera-1",  
  "confidence_threshold": 0.5  
}
```

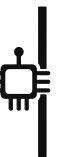


3.2 Software Architecture : VM



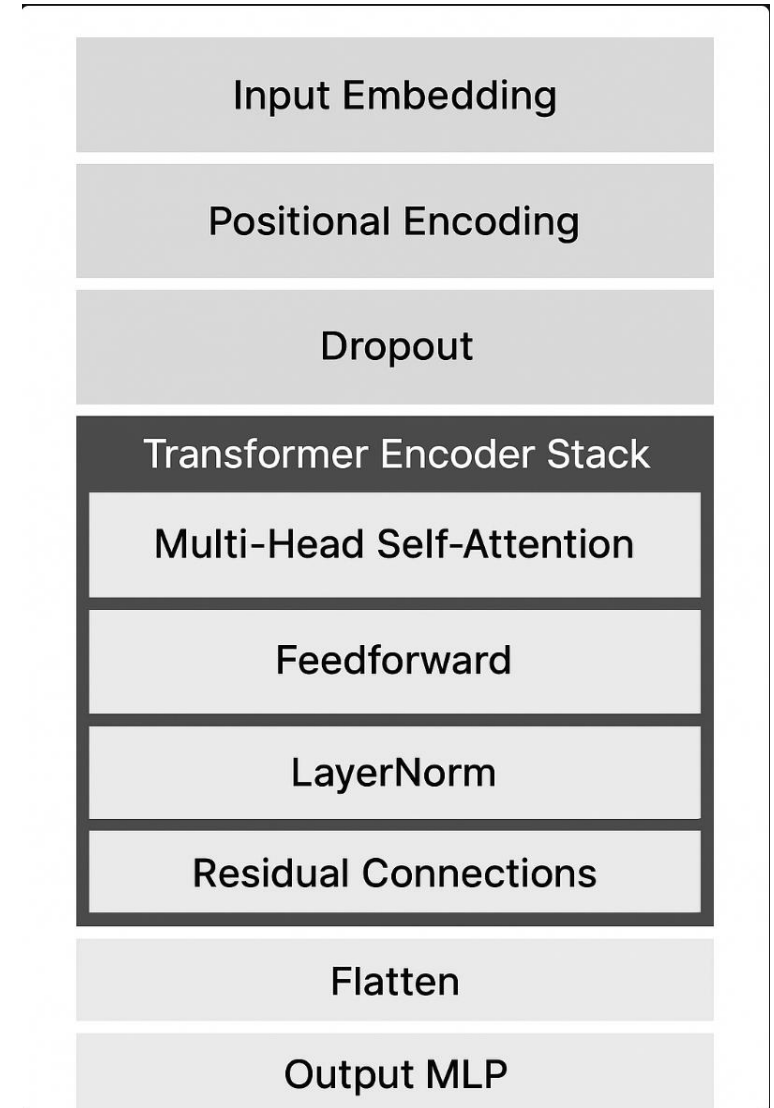
3.3 Transformer model

- Time based (parallel timestamp training)
- RNN -> vanishing gradient (short-term-memory lost over long training)
- Transformer -> **attention weights (gives value on how important the timestamp data is)**
- **"The value of this timestamp 24 hours ago is important for current prediction"**
- Eg.: timestep 1 looks for 50% to timestep 2 to determine it's output



3.3 Transformer model

- PyTorch Encoder-Only Transformer (sequential data)
- Encoder only => classification prediction



Traffic Ops Console

Realtime edge → cloud telemetry

Server: 12/12/2025, 15:59:59
Last refresh: 12/12/2025, 15:59:59

Monitoring

Predictions

Last 10 observations

camera · rock5

Timestamp	Total	Split	Snapshot
12/12/2025, 15:59:49	0 voertuigen	–	View
10/12/2025, 21:15:10	0 voertuigen	–	–
10/12/2025, 21:10:35	2 voertuigen	1 car	–
10/12/2025, 21:03:33	5 voertuigen	4 car	–
10/12/2025, 20:57:17	7 voertuigen	6 car	–
10/12/2025, 20:51:13	2 voertuigen	1 car	–
10/12/2025, 20:49:04	2 voertuigen	1 car	–
10/12/2025, 20:32:59	2 voertuigen	2 car	–
10/12/2025, 20:21:33	3 voertuigen	3 car	–
10/12/2025, 20:15:07	8 voertuigen	6 car	–

Sampling: 10s · auto-refresh every 5s

System status

edge · backend · db

Edge node

● Rock5 camera pipeline

State **ONLINE** ●
Δt sample: 9s

Core API

● FastAPI service

State **ONLINE** ●
DB: OK

Live link OK · polling /api/v1/status

Traffic Ops Console

Realtime edge → cloud telemetry

Server: 12/12/2025, 16:15:04
Last refresh: 12/12/2025, 16:15:04

Monitoring

Predictions

Last 10 observations

Timestamp

12/12/2025, 16:14:28

12/12/2025, 16:14:18

12/12/2025, 16:14:08

12/12/2025, 16:13:57

12/12/2025, 16:00:20

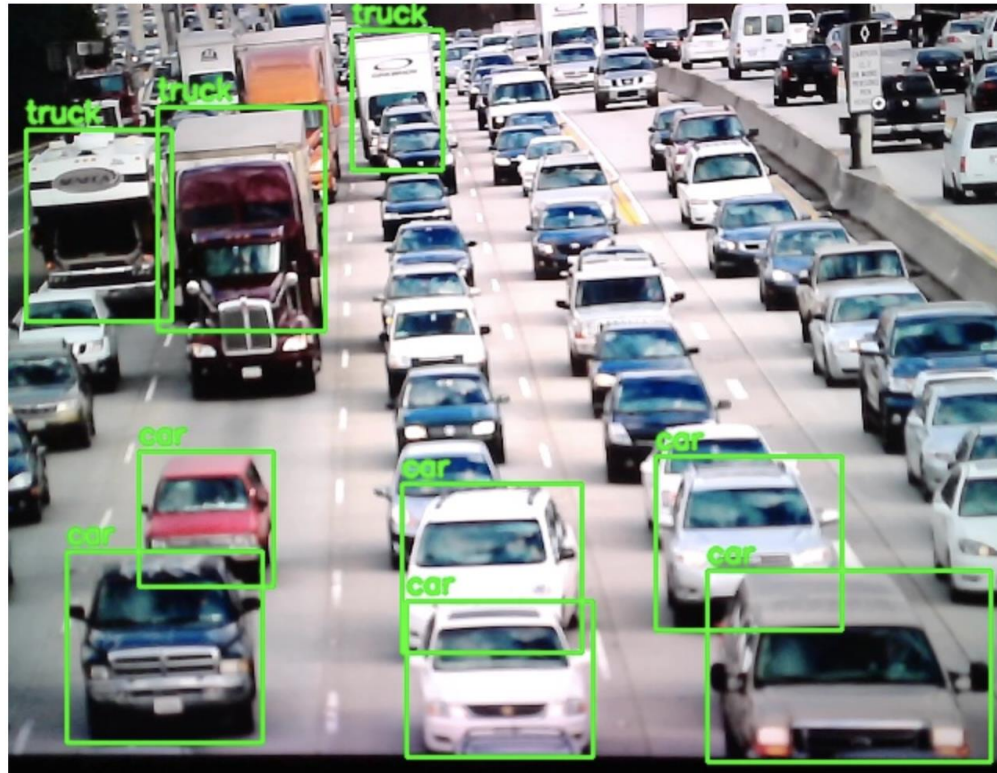
12/12/2025, 16:00:09

12/12/2025, 15:59:59

12/12/2025, 15:59:49

10/12/2025, 21:15:10

10/12/2025, 21:10:35



Sampling: 10s · auto-refresh every 5s

Live link OK · polling /api/v1/status

edge · backend · db

State LAGGING ●

Δt sample: 36s

State ONLINE ●

DB: OK

baseline

Traffic Ops Console

Realtime edge → cloud telemetry

Server: 12/12/2025, 16:00:09
Last refresh: 12/12/2025, 16:00:09

Monitoring

Predictions

Traffic Forecast

AI Transformer

Today

Tomorrow

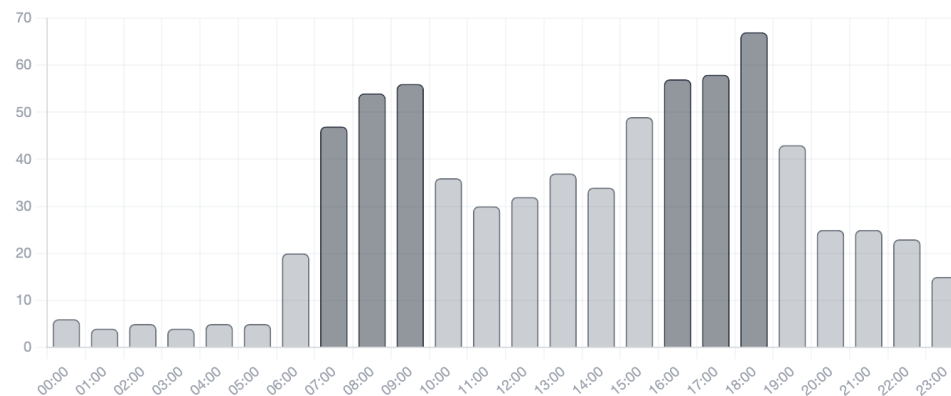
Sun, Dec 14

Mon, Dec 15

Tue, Dec 16

Wed, Dec 17

Thu, Dec 18



Transformer model active

Current Hour

live

PREDICTED VEHICLES

57

cars this hour

RUSH HOUR TRAFFIC

PEAK HOUR

18:00

PEAK CARS

67

DAILY TOTAL

737

RUSH HOUR AVG

56.5

00:00

6 cars

01:00

4 cars

02:00

5 cars

03:00

4 cars

VISIONLANE

Full AI-Traffic detection/prediction

